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Cost of Explainability in AI: an Example with Credit Scoring Models

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ABSTRACT

This paper examines the **cost of explainability in machine learning models for credit scoring**.

The analysis is conducted under the constraint of meeting the regulatory requirements of the European Central Bank (ECB), using a real-life dataset of over 50,000 one-year credit exposures.

We compare the statistical and financial performances of black-box models, such as XGBoost and neural networks, with inherently explainable models like logistic regression and GAMs. Notably, statistical performance does not necessarily correlate with financial performance.

Our results reveal a **difference of 15 to 20 basis points in annual return on investment** between the best performing black-box model and the best performing inherently explainable model, as cost of explainability. We also find that the cost of explainability increases together with the risk appetite.

To enhance the interpretability of explainable models, we apply isotonic smoothing of features' shape functions based on expert judgment. Our findings suggest that **incorporating expert judgment** in the form of isotonic smoothing **improves the explainability without compromising the performance**.

These results have significant implications for the use of explainable models in credit risk assessment and for regulatory compliance.

***NB.** In this presentation, when we cite some authors, we refer to the References available in the original article to be published in the Proceedings of the Conference (open access)*

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Introduction

Data

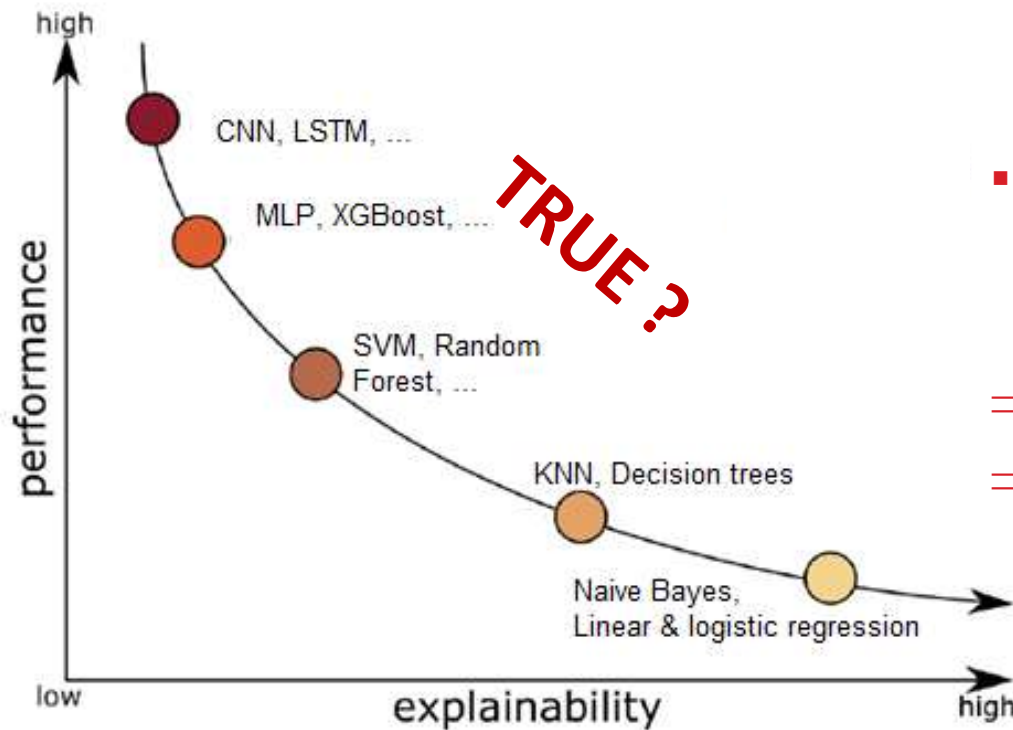
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IS THERE A TRADE-OFF BETWEEN PERFORMANCE AND EXPLAINABILITY ?



- Popular wisdom expects a **trade-off between performance and explainability**

⇒ Is it true ?

⇒ How to measure explainability and performance ?

- Difference between :

- **White box: “inherently explainable”*** statistical inference models (linear and logistic regressions, Naïve Bayes and more generally GLM** and GAM)
- **Black box “ex-post interpretable”** algorithms can benefit from ex-post local explanatory techniques (neural networks, complex decision trees, SVM, etc)

* For this presentation, we consider “inherently explainable” and “intrinsically explainable” as synonyms)

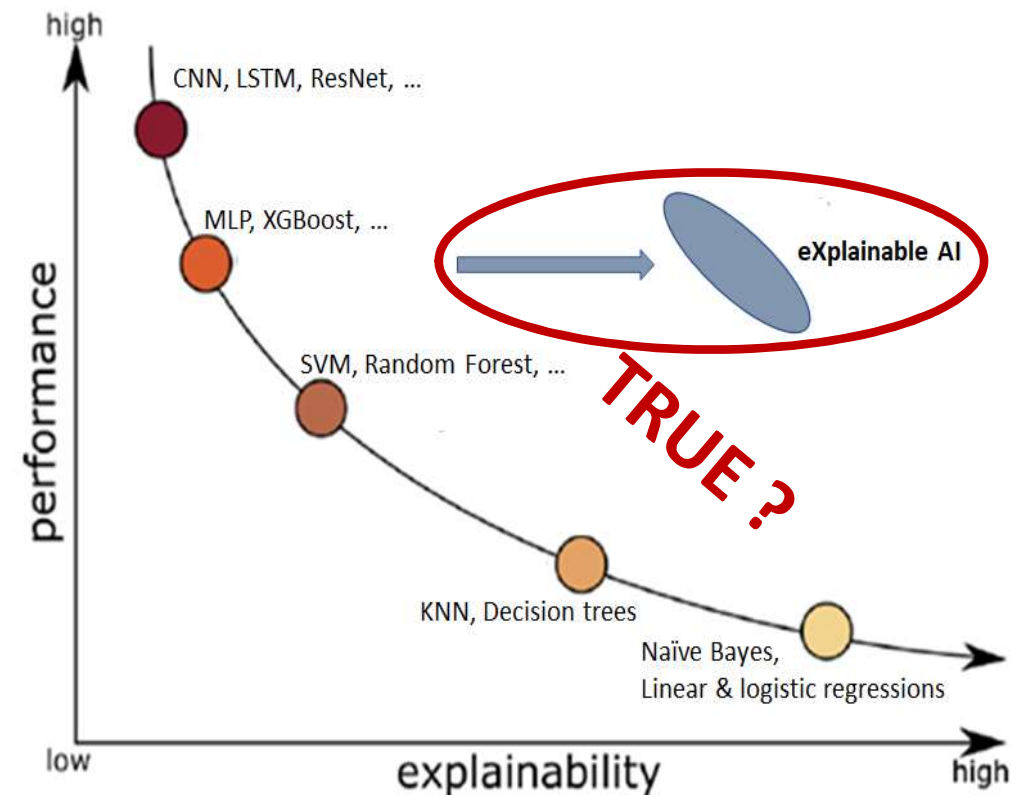
** Abbreviations and model names are further briefly described in **Appendix 1 - Glossary**

IF THERE IS A TRADE-OFF, CAN EXPLAINABLE AI DISRUPT IT ?

- Recent “**eXplainable AI**” algorithms (“**XAI**”):
 - Rely on black-box algorithms (XGB, MLP, ...)
 - Produce a GAM

⇒ Inherently explainable GAM

⇒ As powerful as black-box algorithm ?



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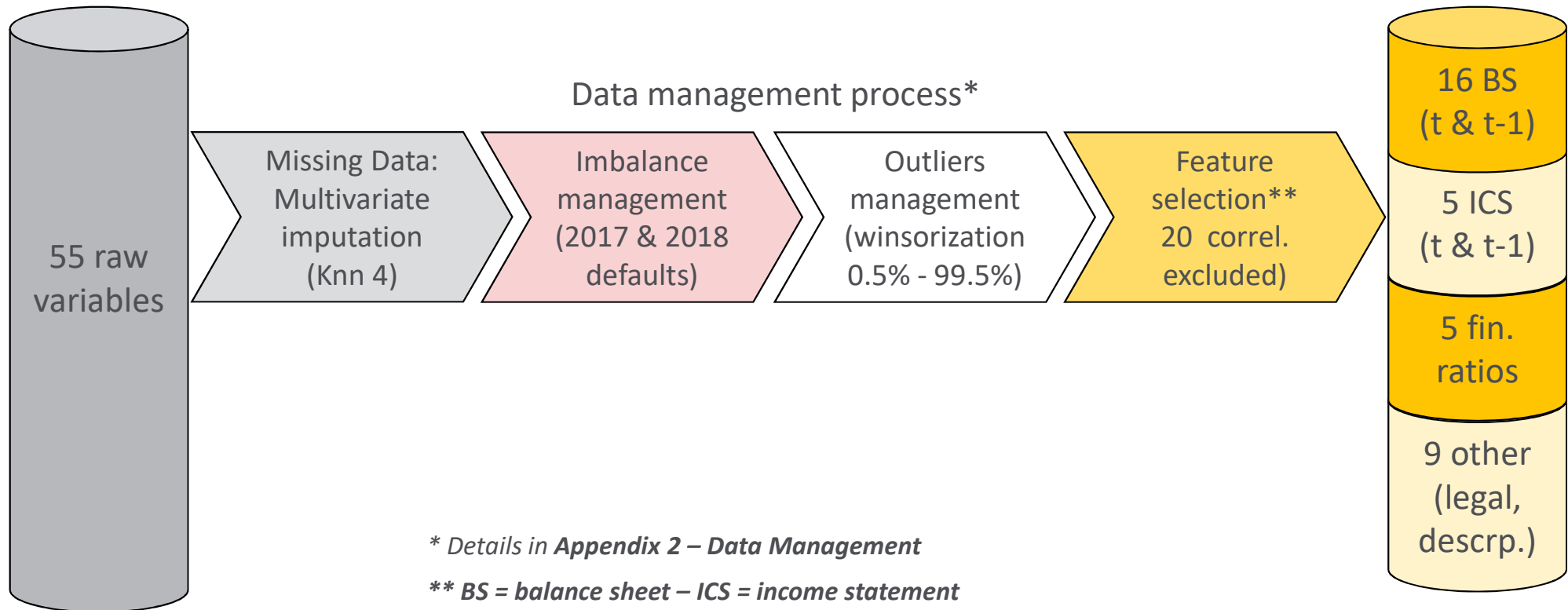
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WE USE REAL-LIFE 1-YEAR CORPORATE CREDIT DATA

- Anonymized **real-life 1-year corporate credit exposures**, with 55 explanatory variables
(*confidential and proprietary dataset*)

	Period	# Obs	# Defaults	Default rate
Training set	2019	76,089	608	0.80%
Enhanced training set	2019++	77,435	1,954	2.52%
Test set 1	2020	44,151	582	1.32%
Test set 2	2021	61,406	335	0.55%
Test set 3	2022	59,074	275	0.47%



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14 MODELS, FROM WHITE-BOX TO BLACK-BOX, ARE TESTED

Models

- **14 models* in total** (among the most common):
 - 4 inherently explainable models
 - 2 explainable AI models
 - Isotonic versions of the 2 explainable AI models, based on expert judgment to “force” the shape of the GAMs
 - 6 black-box models whose interpretation can be done locally and ex-post
- Hyper-parameters tuning with a 2-step grid search
- **Reproducibility and replicability** guaranteed with several measures** (seed, deterministic CUDNN, ...)

Model	Abbrev.	Type	Expert judgment(*)	
Logistic regression	LR	IE	No	Explainable
ElasticNet	EL	IE	No	
Naïve Bayes	NB	IE	No	
Linear Discriminant Analysis	LDA	IE	No	
<u>Explainable Boosting Machine</u>	EBM	XAI	Yes	
<u>GamiNet</u>	GAMI	XAI	Yes	Black-box
Support Vector Machine	SVM	BB	No	
Random Forest	RF	BB	No	
Gradient Boosting	GB	BB	No	
eXtreme Gradient Boosting	XGB	BB	No	
Light GBM	LGBM	BB	No	
Multi-Layer Perceptron	MLP	BB	No	

IE = inherently explainable

XAI = explainable AI, classified as inherently explainable

BB = Black-box, interpretable locally and ex-post

(*) Isotonicity of the GAM forced according to financial expert judgment

* The format of the data does not allow to apply more complex neural networks as CNN, LSTM or ResNet. While these models tend to overperform MLPs, they would require longer historical data for a reliable analysis

Model calibration and regulatory testing

- **Model Calibration:**
 - Each model **generates a PD between 0% and 100%** for each data
 - PDs converted into grades with a **masterscale of 10 grades** normally distributed (following BIS IFC paper)

- **Regulatory requirements:** exclusion of models not matching the ECB regulatory requirements* :
 - Grades: minimum 8 grades, of which 7 non-defaulted, with differences between thresholds > 0.001%
 - Predictive ability: Jeffrey's test to assess the prudence of PD estimates
 - Discriminatory power: tested with AUROC with Mann-Whitney U statistic
 - Stability:
 - Grade dispersion with Herfindal Index (HI)
 - Borrower migrations with matrix weighted bandwidth (MWB) analysis
 - Monotonicity off-diagonal transition frequencies with pairwise z-tests
 - Explainability: **No explicit test from ECB** but general requirement and topic of the analysis

* ECB "Instructions for reporting the validation results of internal models - IRB Pillar I models for credit risk", 2019

FINANCIAL RESULTS ARE USED TO ASSESS MODELS' PERFORMANCE

Financial Performance analysis

- Financial data:

Financial parameter	Abbrev.	Value	Comment
Risk-free rate	rfr	3.25%	1-year government bond yield
Fund-transfer pricing	ftp	0.75%	for funding & liquidity costs, set by ALM
Credit spread	csi	model-based	capped per grade at reference masterscale's PD* LGD
Commercial margin	cm	0.50%	to remunerate commercial and BO departments and capital
Loss-given default	LGD	45.00%	IRB value for senior unsecured credit to corporates in foundation IRB (CRE 32.5)

- Actual **financial result per accepted loan for each model** is equal to:
 - Paid-back: $cs_i + cm$ = credit spread + commercial margin
 - Defaulted: $-(1 + rfr + ftp + cs_i + cm) * LGD$ = total exposure at risk * the LGD

- Risk appetite threshold: based on the grades provided by the algorithms with:
 - low threshold : below grade 6 => lot of loans rejected as too risky
 - Base case: threshold for acceptance set at grade 6
 - High threshold : grades 7 or 8 => most loans accepted, only limited rejections

* 10 grades reference masterscale:

Masterscale	0	1	2	3	4	5	6	7	8	9
PD	0.00%	0.01%	0.02%	0.03%	0.18%	1.15%	4.33%	13.73%	32.95%	100.00%

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NOT ALL MODELS ARE ROBUST ENOUGH TO MATCH ECB REQUIREMENTS

- 11 out of 14 models succeed with ECB requirements
- Naïve Bayes, SVM and Random Forest fail to provide at least 7 non-defaulted grades and are excluded from the analysis
- LDA delivers weak results for the predictive ability but is kept

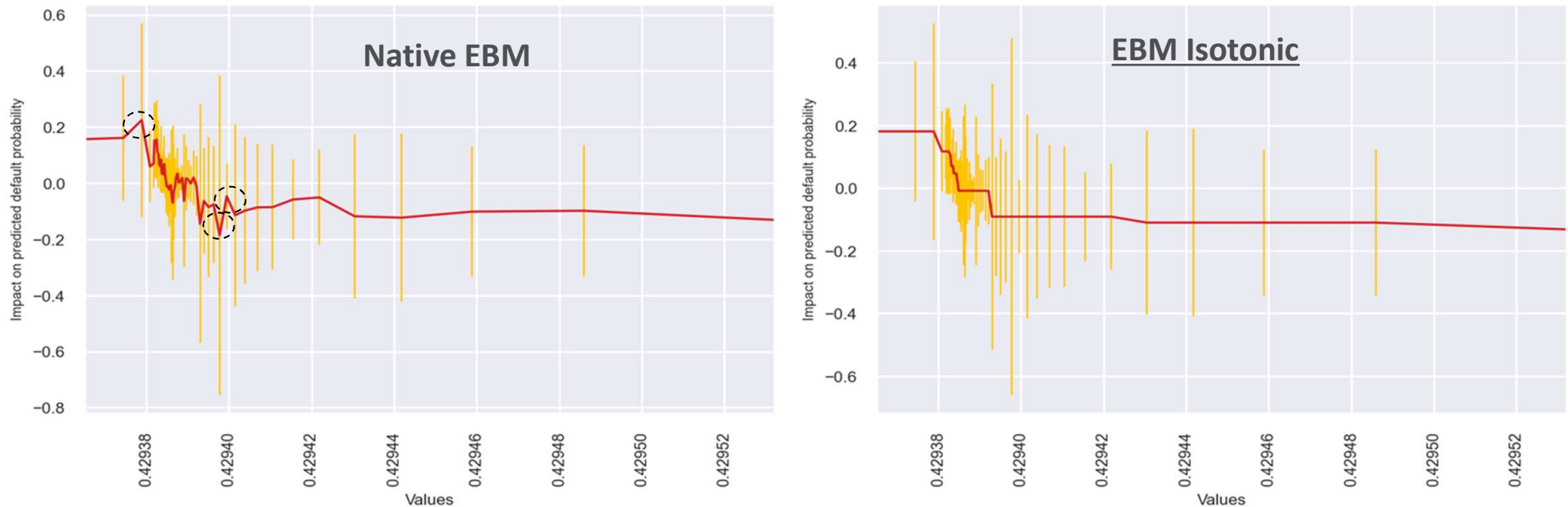
Model	# grades	Predictive ability	Discriminat. power	Stability*	Explainability
Logistic regression	Yes	Yes	Yes	Yes	Yes
ElasticNet	Yes	Yes	Yes	Yes	Yes
Naïve Bayes	No	NR	NR	NR	Yes
Linear Discriminant Analysis	Yes	Weak	Yes	Yes	Yes
Explainable Boosting Machine	Yes	Yes	Yes	Yes	Yes
GamiNet	Yes	Yes	Yes	Yes	Yes
Isotonic EBM	Yes	Yes	Yes	Yes	Yes +
Isotonic GAM	Yes	Yes	Yes	Yes	Yes +
Support Vector Machine	No	NR	NR	NR	NR
Random Forest	No	NR	NR	NR	NR
Gradient Boosting	Yes	Yes	Yes	Yes	NR
eXtreme Gradient Boosting	Yes	Yes	Yes	Yes	NR
Light GBM	Yes	Yes	Yes	Yes	NR
Multi-Layer Perceptron	Yes	Yes	Yes	Yes	NR

* Stability has been tested on a small sample (2 transitions) during an unusual period marked by covid-19
All algorithms succeed with HI test and MWB, but most face minor issues with z-tests

- All other models succeed with the ECB tests
- General requirement but **no specific ECB test for the explainability**

EXPLAINABLE MODELS CAN BE IMPROVED WITH EXPERT JUDGMENT

- **Expert judgment improves the explainability** of EBM and GAMI, making the GAM-shapes easier to understand. Example with bs_012 feature (normalized amount of retained earnings) for which a monotonic negative correlation with the PD is expected:



- Monotonizing the shape of the GAMs:
 - ⇒ **Increased explainability** with a GAM shape that follows market expectation
 - ⇒ **Reduced risk of overfitting**
 - ⇒ **Possible outliers** impact eliminated with isotonic model

ALL MODELS ADD VALUE, SOME MUCH MORE THAN OTHERS...

Financial results 2020

- All models significantly improve the credit portfolio management compared to a “no-model” situation
- GAMI is closely followed by EBM as best performer for inherently explainable model, far ahead of the 3 other “standard” models
- **MLP significantly over-performs** the other black-box models
- **Isotonic XAI models** that integrate expert judgment **come at virtually no cost**, compared to native XAI models
- Risk appetite impacts the rejection rate. Therefore, low risk appetite might create commercial issues

Model	Threshold Grade 6		Threshold Grade 7		
	Accepted	Return	Accepted	Return	
LR	88.53%	1.34%	97.76%	1.52%	Explainable
ELN	88.67%	1.31%	97.60%	1.49%	
LDA	88.30%	1.20%	97.98%	1.38%	
EBM	89.09%	1.46%	97.56%	1.64%	
<i>EBM Isotonic</i>	89.11%	1.46%	97.46%	1.65%	
GAMI	88.45%	1.47%	96.84%	1.66%	
<i>GAMI Isotonic</i>	88.28%	1.45%	96.88%	1.64%	Black-box
XGB	88.78%	1.43%	97.55%	1.60%	
LGBM	87.44%	1.08%	97.43%	1.31%	
GB	88.16%	1.38%	96.61%	1.57%	
MLP	88.33%	1.61%	96.95%	1.83%	
<i>No model</i>	100.00%	0.55%	100.00%	0.55%	

Threshold	Grade 4	Grade 5	Grade 6	Grade 7	Grade 8
Rejection	65.3%	30.9%	9.1%	1.8%	0.2%

- 2021 and 2022 results are very similar. Please refer to corresponding author for further details

TRADE-OFF BETWEEN EXPLAINABILITY AND PERFORMANCE EXISTS.. BUT EXPLAINABLE AI REDUCES THE COST OF EXPLAINABILITY

- From the analysis for a risk appetite threshold at grade 6 and 7 respectively, we can deduct a cost of explainability :

		2020	2021	2022
Low risk (threshold 6)	Best XAI model	1.47%	1.66%	1.57%
	Best model	1.61%	1.82%	1.77%
	Cost of XAI	0.14%	0.16%	0.20%
High risk (threshold 7)	Best XAI model	1.66%	1.88%	1.73%
	Best model	1.83%	2.06%	1.94%
	Cost of XAI	0.17%	0.18%	0.21%

- The **cost of explainability**:
 - Fluctuates in a range of **15 to 20bp return on investment (RoI)**
 - Increases towards 25 to 30 bp at least with traditional white box models
- The **purpose of the model** (commercial for pricing and underwriting, risk management, capital consumption, ...) might **influence the importance of the cost of explainability** and should drive the preference for the best model or for the best explainable model, rather than for traditional explainable algorithms

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Conclusion

- This paper examines the **cost of explainability in machine learning models for credit scoring** from a financial perspective, using real-life corporate credit exposures
- We compare two types of models: traditional models inherently explainable and new XAI models generating a GAM versus ex-post interpretable black-box models. Based on our dataset, we find that:
 - **Several standard models do not match ECB requirements** and are excluded from the comparison
 - **Best black-box** (neural network) **outperforms** the best explainable model by a **0.15% to 0.20% return on investment (RoI)**
 - The cost of explainability increases to 0.30% when only traditional inherently explainable models like logistic regression are considered (XAI models excluded)
 - Best XAI models allow for **expert judgment that increases explainability at virtually no cost** in terms of RoI
- If confirmed with other datasets, this can have and should have **significant impact** in terms of:
 - **Model acceptability from a regulatory perspective** for calculating regulatory capital requirements
 - **Model choice** by financial institutions for the various tasks (underwriting and pricing, credit risk management, regulatory capital, etc)

THIS PROMISING RESULT SHOULD BE CONFIRMED AND EXTENDED

Perspectives

- The proposed approach shows promising results in terms of cost of explainability
- These results should be confirmed with:
 - **More granular grading** and more granular risk appetite thresholds
 - **Different datasets:**
 - Longer periods that would allow the use of more complex neural networks with 3D input tensors that capture the time-series dimension (LSTM, CNN, ResNet, ...)
 - Longer exposure at risk than the 1-year credit risk, with the increased complexity of 1-year PD not equal to lifetime PD
 - Different types of exposures (mortgages, consumer loans, P2P, specialized finance, micro-finance, ...)
- Discussions with **regulators regarding the acceptability of XAI** models like EBM or GAMI is a prerequisite for a larger diffusion of these powerful explainable models
- Furthermore, **the use of XAI models** like EBM or GAMI could improve:
 - **Data management and outliers handling** through expert judgment
 - The **feature selection process**, relying on the assessment of the importance and the impact of each feature in the GAM

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APPENDIX 1 - GLOSSARY



Term	Type	Explanation
AUC	Metric	Area Under the Curve (AUC) is a performance metric that represents the aggregate model performance.
AUROC	Metric	Area Under the Receiver Operating Characteristic (AUROC) is a measure of classification accuracy.
BIS	Organization	Bank for International Settlements (BIS) is an international financial institution that serves as a bank for central banks and promotes monetary and financial stability worldwide.
CNN	Model	Convolutional Neural Network (CNN) is a deep learning architecture commonly used for image analysis.
CPU	Hardware	Central Processing Unit (CPU) is the primary component responsible for executing computer programs.
CUDNN	Software	CUDA Deep Neural Network Library (CUDNN) is a GPU-accelerated library for deep neural networks.
EBM	Model	Explainable Boosting Machine (EBM) is a model that provides interpretable explanations for predictions.
ECB	Organization	European Central Bank (ECB) is the central bank for the euro and administers monetary policy.
ElasticNet	Model	ElasticNet is a linear regression model that combines L1 and L2 regularization techniques.
GAM	Model	Generalized Additive Model (GAM) is a flexible regression model that incorporates non-linearities.
GAMI	Model	GamiNet is a model that provides interpretable explanations for predictions.
GB	Model	Gradient Boosting (GB) is an ensemble learning technique that builds models in a sequential manner.
GLM	Model	Generalized Linear Model (GLM) is a flexible regression model that can handle various distributions.
GPU	Hardware	Graphics Processing Unit (GPU) is a specialized hardware that accelerates graphics and computations.
HI	Metric	Herfindahl Index (HI) is a measure of market concentration used to assess the concentration of grades
ICE	Plot	Individual Conditional Expectation (ICE) plot visualizes the relationship between a feature and target variable at an individual level.
IFC	Model	Independent Factor Contribution (IFC) measures the contribution of a predictor in a model.
IRB	Regulation	Internal Ratings-Based (IRB) approach is a method for calculating credit risk under Basel II framework.
JS	Metric	Jensen-Shannon Divergence (JS) measures the similarity between two probability distributions.
KL	Metric	Kullback-Leibler Divergence (KL) measures the difference between two probability distributions.
KNN	Model	k-Nearest Neighbors (KNN) is a non-parametric algorithm that classifies data based on proximity.
KS	Metric	Kolmogorov-Smirnov (KS) statistic measures the maximum difference between two cumulative distribution functions.
Lasso	Model	Lasso (Least Absolute Shrinkage and Selection Operator) is a linear regression model with L1 regularization.
LDA	Model	Linear Discriminant Analysis (LDA) is a classification technique that finds linear combinations of features.
LGBM	Model	Light Gradient Boosting Machine (LGBM) is a gradient boosting framework known for its speed and efficiency.
LGD	Metric	Loss Given Default (LGD) measures the amount of loss incurred by a lender in the event of a default.

APPENDIX 1 - GLOSSARY



Term	Type	Explanation
LIME	Model Interpretation	Local Interpretable Model-agnostic Explanations (LIME) provides interpretability for black-box models.
LR	Model	Logistic Regression (LR) is a statistical model used to predict the probability of a binary or categorical outcome based on input variables. It models the relationship between the predictors and the log-odds of the target variable using a logistic function.
LSTM	Model	Long Short-Term Memory (LSTM) is a type of recurrent neural network (RNN) architecture.
Masterscale	Metric	Masterscale is the table that associates a PD to various ratings of grades in a monotonic increasing order.
MLP	Model	Multilayer Perceptron (MLP) is a type of artificial neural network with multiple hidden layers.
MWB	Method	Matrix Weighted Bandwidth (MWB) is a technique used in kernel density estimation to determine the optimal bandwidth for smoothing data based on a weighted matrix.
PD	Metric	Probability of Default (PD) is the likelihood of a borrower defaulting on a loan or credit obligation.
PDP	Plot	Partial Dependence Plot (PDP) shows the relationship between a feature and the predicted outcome, while holding other features constant.
PSI	Metric	Population Stability Index (PSI) measures the distributional changes in a variable over time.
ResNet	Model	Residual Network (ResNet) is a deep learning architecture that introduces skip connections to overcome vanishing gradient problems.
RF	Model	Random Forest (RF) is an ensemble learning method that combines multiple decision trees for classification or regression.
ROC	Metric	Receiver Operating Characteristic (ROC) curve visualizes the trade-off between true positive rate and false positive rate.
RoE	Metric	Return on Equity (RoE) measures the profitability relative to shareholders' equity.
RoI	Metric	Return on Investment (RoI) measures the profitability of an investment relative to its cost.
SHAP	Model Interpretation	Method using Shapley Values, which provide an explanation for individual predictions in machine learning models.
SVM	Model	Support Vector Machine (SVM) is a supervised learning algorithm used for classification and regression tasks.
Winsorization	Data Transformation	Winsorization is a technique that replaces extreme values in a dataset with less extreme values.
XGB	Model	XGBoost (eXtreme Gradient Boosting) is an optimized gradient boosting framework.

APPENDIX 2 – DATA MANAGEMENT

Additional information related to data management

■ Missing data:

- Some algorithms can cope with missing data, most don't. Therefore, we used imputation techniques
 - We tested several imputation methods:
 - Zero value
 - Median (univariate)
 - Bayesian-ridge with round-robin iterative process (multivariate)
 - **K-nearest Neighbors KNN** (multivariate) with various values for k.
- ⇒ KNN with k=4 is **best performing method** following PSI, Kolmogorov-Smirnov (KS), Kullback-Leibler (KL) divergence and Jensen-Shannon (JS) divergence tests

■ **Standardization or min-max normalization:** Following most common practice in data science, we apply:

- Standardization: Logistic Regression, ElasticNet, LDA, Naïve Bayes, Random Forest and SVM
- Normalization: GB, XGB, LGBM, EBM, GAMI and MLP

We also tested the normalization with the models that used standardization for sanity check

APPENDIX 3 – HARDWARE & SOFTWARE

Hardware :

In order to assess the reproducibility of the models, two different PCs were used to perform the computations

HARDWARE	PC1 (main)	PC2 (check)
Processor	AMD Ryzen 9 3950 X	Intel i7-8700
CPU	DDR4	DDR3
GPU	Nvidia RTX 3090 24GB	Nvidia GTX 1060i 6GB
RAM	64 GB	16 GB

Software :

The two computers run similar software with, sometimes, marginal differences in releases

SOFTWARE	PC1 (main)	PC2 (check)
OS	Windows 10 Pro	Windows 10 Pro
Anaconda	2.4.2	2.4.2
Spyder	5.4.3	5.3.3
Python	3.10.8	3.9
Cuda	11.7.1	11.3.1
CUDNN	8.2.1	7.6.5
Pytorch	1.13.1	1.12.1
scikit-learn	1.0.2	1.0.1
XGBoost	1.5.0	1.5.0
Lightgbm	3.2.1	3.2.1
Interpret	0.3.0	0.3.0
Gaminet	1.0.0	1.0.0
Numpy	1.21.5	1.21.5
Pandas	1.4.4	1.4.2
Matplotlib	3.6.2	3.6.2

In order to secure reproducibility to the larger extent possible, we applied several strategies :

- Seed defined for Python, Numpy and Pytorch (both CPU and GPU)
- Deterministic backend forced for CUDNN
- debug environment variable CUBLAS_WORKSPACE_CONFIG defined to ":4096:8"

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Consult our **Credit risk practice** section